A lot of my recent work has involved asking a fairly simple question: when should we trust an AI system, and how do we know when we should not? A few years ago, I wrote an article discussing how historical biases in healthcare research could easily be amplified by AI systems that lack the ability to second-guess themselves. On my first day of master's classes, I reached out to a professor at BMC working in bioethics to learn more about healthcare AI research in Uppsala. Later, we attended an AI Act conference through the healthcare lens and worked on a grant proposal investigating why patients are often more forgiving of a doctor making a mistake than an AI. I also spent time exploring my own idea for a 1177-supportive nurse AI, which eventually made me realise how difficult reliability, regulation, and uncertainty become once these systems move toward real healthcare settings.

That question has stayed with me during my master's thesis on AI-assisted moderation of social media comments under the Digital Services Act. The technical side of LLMs has advanced very quickly, yet uncertainty and reliability still feel like the real bottlenecks for deployment in sensitive domains. That is what made this PhD particularly interesting to me. The focus on uncertainty quantification and probabilistic modelling around LLM outputs feels much closer to the actual problems healthcare systems need solved than simply improving benchmark performance.

Growing up in a hospital compound where my father worked gave me early exposure to healthcare systems and the reality that clinical information is often messy, inconsistent, and context-dependent. During my master's in Data Science at Uppsala, I deliberately moved toward healthcare AI and health data systems. In a recent collaboration with a healthtech company working on EHR interoperability, I interviewed clinicians and hospital IT staff about how healthcare data is actually used in practice.

On the technical side, I am very comfortable with Python and machine learning workflows. My thesis involves transformer-based NLP pipelines, fine-tuning models such as RoBERTa and HateBERT, few-shot prompting with LLMs, and designing annotation frameworks for ambiguous classification tasks. While my background is more applied than theoretical, I do have a solid mathematical foundation from both my software engineering degree and quantitative work during my master's. I would not claim deep expertise in Bayesian statistics yet, but probabilistic modelling and uncertainty estimation are exactly the directions I want to develop further through doctoral research.
What also attracts me to this position is the combination of methodological research with real clinical collaboration. I am most motivated by research that becomes useful to clinicians, researchers, or policymakers outside the university itself. The earliest I would be ready to start is the start of July, assuming my thesis gets graded on time and I receive my degree.